SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.2: Current Electricity — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Current Electricity
Driving Question	DRIVING QUESTION: Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon: The Huruma Estate Mystery — Why Are the Lights Dim and the Kettle Slow?**	Students observed a scenario (photo/video clip or teacher narration) of a Huruma Estate household experiencing dim lights and a slow-boiling electric kettle immediately after power restoration following a Kenya Power outage. They noted that some appliances appeared to work normally while others did not, even though all were plugged in and switched on. Students recorded initial observations and generated questions on sticky notes for the Driving Question Board (DQB).	Students learned that electrical devices do not always receive the energy they are designed to use, and that the circuit conditions — not just the power supply — determine how much energy each device actually gets. They were introduced to the sub-strand's anchoring phenomenon and the Driving Question. Key vocabulary introduced: electric current, voltage (potential difference), resistance, circuit.	Pedagogical note: Do NOT resolve the phenomenon here. The teacher's role is to sustain productive confusion and curiosity. Accept all student explanations without evaluating correctness — record every idea on the DQB. Cold-call at least 6 students using equity sticks. Prompt deeper thinking with: 'What information would you need to figure out why the kettle is slow?' This lesson establishes the phenomenon that every subsequent lesson will build toward explaining.
Lesson 2 **Electric Current and Potential Difference: The Flow of Charge and the 'Push' That Drives It**	Students observed a simple demonstration circuit (battery, bulb, connecting wires, ammeter, voltmeter) built by the teacher using locally available components. They noted the ammeter reading changed when an extra battery cell was added, and that the voltmeter measured the 'push' across the bulb. Students also examined a diagram of conventional current direction versus electron flow direction and completed a structured observation table.	Students learned that: (1) electric current (I) is the rate of flow of charge, defined as $I = Q/t$, measured in amperes (A); (2) potential difference/voltage (V) is the work done per unit charge to move charge between two points, measured in volts (V); (3) conventional current flows from positive to negative terminal externally, while electrons flow in the opposite direction. These two quantities — current and voltage — are the foundation for understanding energy delivery in every circuit in the sub-strand.	Pedagogical note: Anchor the explanation back to Huruma Estate — 'If the voltage (push) from Kenya Power is lower than expected after restoration, how would that affect the current through the kettle's heating element?' Allow 30 seconds of wait time before cold-calling. Emphasise the analogy: voltage is like water pressure in a Nairobi pipe network; current is like the flow rate of water. Students should update their DQB with new understanding: 'Low voltage → less current → less energy delivered to appliance.'

Lesson 3 **Ohm's Law Investigation: Resistance as the Controller of Current**	Students conducted a hands-on investigation varying voltage across a resistor (or a length of nichrome wire) and measuring the resulting current using an ammeter. They plotted a V–I graph and observed that, for an ohmic conductor, the graph is a straight line through the origin. They also tested a filament bulb and observed the non-linear (curved) V–I graph, noting that its resistance increases as it heats up.	Students learned that the gradient of a V–I graph gives the resistance (R) of the conductor, measured in ohms (Ω). This relationship is summarised by Ohm's Law: $V = IR$, which holds for ohmic conductors at constant temperature. Non-ohmic conductors (e.g., filament bulbs, diodes) do not produce a straight-line V–I graph. Resistance is the opposition to the flow of current — a higher resistance means less current for the same voltage.	Pedagogical note: Connect directly to the DQ — 'If there is extra resistance somewhere in the Huruma Estate circuit, what does Ohm's Law predict will happen to the current, and therefore to the energy the kettle receives?' The V–I graph gradient skill is assessable; ensure every student calculates R from their own graph. Use the Kenyan context: resistance in household wiring is like a narrow matatu route — the same number of passengers (charge) struggle to flow through. Collect and check graphs before the lesson ends.
Lesson 4 **Resistance and Resistivity: Why Do Thicker Cables Carry More Current?**	Students compared the resistance of wires of different lengths, thicknesses, and materials (e.g., copper vs. nichrome) by measuring current at constant voltage and applying Ohm's Law ($R = V/I$). They observed that longer wires produced lower currents (higher resistance), thicker wires produced higher currents (lower resistance), and different materials produced different resistances even at the same dimensions.	Students learned that resistance depends on three factors: length ($R \propto L$), cross-sectional area ($R \propto 1/A$), and material (characterised by resistivity ρ). These are combined in the formula $R = \rho L/A$, where ρ is the resistivity of the material in ohm-metres (Ω·m). Copper has very low resistivity, making it ideal for household wiring. A long, thin wire has much higher resistance than a short, thick wire of the same material.	Pedagogical note: This lesson directly explains one cause of the Huruma phenomenon — long, undersized wiring from the Kenya Power transformer to the estate increases resistance, causing a voltage drop before electricity reaches homes. Ask students: 'Why do electricians in Kenya use thick copper cables for high-current appliances like electric cookers, but thin wires for low-current devices like phone chargers?' Expect students to reference $R = \rho L/A$. Update the DQB: 'Long or thin wiring in the estate adds resistance → voltage drop → devices receive less energy.' Pedagogical caution: distinguish clearly between resistivity (a material property) and resistance (a component property).
Lesson 5 **Series Circuits: Same Current, Sharing Voltage — Why More Bulbs Mean Dimmer Light**	Students built or observed series circuits with 1, 2, and 3 identical bulbs connected to the same battery. They measured voltage across each bulb and current at various points in the circuit. They observed: (a) the ammeter reading was the same at all points; (b) each bulb became dimmer as more bulbs were added; (c) the voltage across each bulb was a fraction of the total supply voltage.	In a series circuit: (1) current is the same everywhere in the loop ($I_{total} = I_1 = I_2 = I_3$); (2) the source voltage is shared (divided) among the components so that $V_{source} = V_1 + V_2 + V_3$; (3) total resistance increases with each added component: $R_{total} = R_1 + R_2 + R_3$. Adding more resistors reduces current and dims bulbs because each component gets less than the full supply voltage.	Pedagogical note: Connect to DQ — 'If Huruma Estate appliances were wired in series (they are not, but hypothetically), adding more devices would dim all of them. Is this what we observe in real homes?' This motivates students to question why real household circuits are NOT wired in series. The teacher should cold-call at least 5 students to explain each of the three series rules in their own words before moving to the Explain phase. Use the githeri analogy: one pot of githeri (voltage) shared equally among the number of people (components) — each person gets less if there are more people sharing.

Lesson 6 **Parallel Circuits: Why Every Appliance in a Kenyan Home Gets Full Voltage**	Students built or observed parallel circuits with bulbs connected across the same two nodes. They measured voltage across each branch and current through each branch and through the main supply. They observed: (a) voltage was identical across every branch; (b) adding branches did not dim the other bulbs; (c) total current from the battery increased with each additional branch; (d) removing one bulb did not affect the others.	Students derived the three rules of parallel circuits: (1) voltage is the same across every branch and equals the supply voltage ($V\_total = V_1 = V_2 = V_3$); (2) total current is the sum of branch currents ($I\_total = I_1 + I_2 + I_3$); (3) the equivalent resistance is less than the smallest branch resistance, found using $1/R\_total = 1/R_1 + 1/R_2 + 1/R_3$. Kenyan household wiring uses parallel circuits so that every appliance receives the full 240 V regardless of how many devices are in use.	Pedagogical note: This is a pivotal lesson for resolving the DQ. Students should now be able to articulate: 'Household circuits are parallel — every appliance should get full voltage. So why are Huruma Estate lights still dim?' This productive tension sets up Lessons 7–8 on internal resistance and voltage drop. Update the DQB accordingly. Reinforce with the Kenya Power analogy: each house on a street is a parallel branch — switching off your neighbour's power does not affect yours. Ensure students can derive R_equivalent for two resistors in parallel using the product-over-sum shortcut: $R = R_1R_2/(R_1+R_2)$.
Lesson 7 **Internal Resistance and Terminal Voltage: Why Batteries and the Grid 'Sag' Under Load**	Students observed (or conducted) an experiment in which a battery's terminal voltage was measured first with no load (open circuit) and then with increasing loads (more bulbs in parallel). They recorded that the terminal voltage dropped as more current was drawn. A graph of terminal voltage (V) vs. current (I) produced a straight line with a negative gradient, with the y-intercept representing the EMF (ε) and the gradient representing the internal resistance (r).	Students learned that every real power source (battery, generator, Kenya Power transformer) has internal resistance (r) that causes the terminal voltage to drop when current flows. The relationship is: $\varepsilon = V\_terminal + Ir$, so $V\_terminal = \varepsilon - Ir$. The 'lost volts' (Ir) are dissipated inside the source. At high current demand (e.g., when many Huruma Estate homes switch on simultaneously after a blackout), the grid's effective internal resistance causes a significant voltage sag, explaining why devices receive less than the rated 240 V.	Pedagogical note: This lesson is the key mechanistic explanation of the anchoring phenomenon. After students derive $V\_terminal = \varepsilon - Ir$, explicitly return to Huruma Estate: 'When everyone in the estate switches on simultaneously after the Kenya Power outage, the sudden surge in current causes the transformer's terminal voltage to drop. This is why lights are dim and the kettle is slow — they are receiving less than 240 V.' Allow students 2 minutes of silent writing to connect this to their DQB before class discussion. Cold-call 4 students to narrate the full causal chain using correct physics vocabulary.
Lesson 8 **Kirchhoff's Laws: Rules for Analysing Complex Circuits**	Students observed a complex circuit (a junction with multiple branches and multiple loops) drawn on the board or provided as a worksheet. Using ammeters and voltmeters (real or simulated), they measured currents at junctions and voltages around loops, then tabulated results. They observed that current 'splitting' at a junction always conserved the total, and that summing voltage gains and drops around any closed loop always returned to zero.	Students learned Kirchhoff's two laws: (1) Kirchhoff's Current Law (KCL) — the algebraic sum of currents at any junction is zero ($\Sigma I\_in = \Sigma I\_out$), a consequence of conservation of charge; (2) Kirchhoff's Voltage Law (KVL) — the algebraic sum of EMFs and potential drops around any closed loop is zero ($\Sigma V = 0$), a consequence of conservation of energy. These laws allow analysis of circuits too complex to simplify using series/parallel rules alone, and underpin professional circuit design used by Kenyan electrical engineers.	Pedagogical note: Kirchhoff's Laws are often introduced abstractly. Ground every example in the Huruma Estate context — e.g., 'Apply KVL around the loop that includes the transformer's internal resistance and the kettle branch. What does this tell us about the voltage the kettle actually receives?' Provide a structured worked example before asking students to solve independently. A common error is inconsistent sign conventions — explicitly model the sign rule (voltage rise = positive, voltage drop = negative) and have students check their answers using KCL at every junction. At least one assessment question should require simultaneous application of

			both laws.
Lesson 9 **Electrical Power: Deriving and Applying $P = IV = I^2R = V^2/R$**	Students observed or conducted an experiment in which they varied current through a resistor (or a small electric heater) and measured both voltage and the rate of temperature rise (as a proxy for power dissipated). They recorded that power increased rapidly with current, and that doubling the current caused the heating rate to quadruple, suggesting a squared relationship. They also read wattage labels on common Kenyan appliances (kettle, jiko la umeme, phone charger, fluorescent bulb).	Students learned that: (1) electrical power $P = IV = I^2R = V^2/R$ is the rate of energy transfer in a circuit, measured in watts (W); (2) power dissipated in a resistor increases with the square of the current ($P = I^2R$), which is why high-resistance components or high-current circuits waste large amounts of energy as heat; (3) a Kenyan household electric kettle rated at 2 000 W at 240 V draws $I = P/V = 2000/240 \approx 8.3$ A; (4) if terminal voltage drops to 200 V (as in Huruma), the kettle only dissipates $P = V^2/R \approx 1\,389$ W — explaining why it heats water more slowly.	Pedagogical note: The calculation in point (4) above is the quantitative resolution of the anchoring phenomenon — ensure every student works through this calculation independently and records it in their DQB model. This is a high-yield lesson for KCSE-style numerical questions. Three forms of the power equation serve different scenarios: use $P = IV$ when both I and V are known; use $P = I^2R$ when I and R are known (e.g., calculating wasted power in a resistive wire); use $P = V^2/R$ when V and R are known (e.g., appliance performance at reduced voltage). Require students to justify formula choice, not just substitute numbers.
Lesson 10 **Electrical Energy: From Watts to Kilowatt-Hours — Reading a Kenya Power Bill**	Students examined a sample Kenya Power electricity bill and identified units consumed (kWh), rate charged per unit, and total cost. They then used a class data set of appliance wattages (from Lesson 9 labels) and estimated usage times to calculate the energy consumed by each appliance per month. They observed that the electric water heater (geyser) and electric cooker dominated household energy consumption.	Students learned that electrical energy $E = Pt$ (energy = power × time), measured in joules (J) for scientific purposes, but billed in kilowatt-hours (kWh) where 1 kWh = 3.6×10^6 J. They calculated monthly energy costs for a typical Kenyan household and identified the highest-consuming appliances. They also learned that reducing voltage (as in the Huruma scenario) reduces power delivered to appliances ($P = V^2/R$), which reduces energy consumed — but also reduces useful output (e.g., slower kettle, dimmer lights), meaning consumers pay for less energy but also receive less service.	Pedagogical note: This lesson bridges physics to real-life financial literacy — a strong PCIs connection (Financial Literacy, Life Skills). Use actual Kenya Power tariff rates (e.g., domestic tariff as published by EPRA) to make calculations authentic. A productive class discussion question: 'If Kenya Power drops voltage from 240 V to 200 V, who benefits — the consumer or the utility? Is this fair?' This engages Values (Social Justice, Responsibility) and can be connected to Kenya's Energy Act. Ensure students can convert between J and kWh fluently, as this is a common KCSE error source.
Lesson 11 **Heating Effect of Electric Current & Electrical Safety**	Students observed demonstrations of Joule heating: a thin nichrome wire glowing red when high current passed through it; a thick copper wire remaining cool under the same current. They also examined (or were shown images of) burnt plug sockets, melted wire insulation, and a tripped circuit breaker — all common in Kenyan homes. A case study of an electrical fire caused by overloaded extension cables in a Nairobi rental apartment was discussed.	Students learned that electrical energy is converted to heat energy in any resistor carrying current, quantified by $Q = I^2Rt$ (Joule's Law). High-resistance conductors or high currents produce dangerous levels of heat. Safety devices exploit or mitigate this effect: (1) fuses — a thin wire of low melting point that melts (breaks the circuit) when current exceeds a safe threshold, rated in amperes; (2) circuit breakers — electromagnetic or thermal switches that trip automatically and can be reset; (3) earthing — provides a low-resistance path to ground to prevent metal appliance	Pedagogical note: Safety content must be taught with genuine urgency — electrical accidents are a leading cause of household fires in Kenya. Use $Q = I^2Rt$ to show quantitatively why a 13 A fuse protects a 2 000 W kettle ($I_{operating} \approx 8.3$ A, fuse provides margin) but would NOT protect a phone charger circuit if multiple high-power devices are added on one extension. The Huruma Estate scenario can be extended: 'When voltage is restored to full after a sag, the sudden current surge can cause heating in undersized wiring — this is a real fire risk.' Connect to DQB: efficient circuit

		casings from becoming live. Correct fuse rating selection (I_{fuse} slightly above normal operating current) is a key design skill.	design includes not just delivering the right voltage but also protecting against dangerous heating.
Lesson 12 **Model Building, DQB Resolution & Final Explanation: Designing Safe and Efficient Circuits for Kenyan Homes**	Students reviewed all evidence gathered across Lessons 1–11: V–I graphs, circuit measurements, power calculations, energy billing data, and safety case studies. They revisited their initial Lesson 1 DQB predictions and compared them with their current understanding. Working in groups, they constructed a final annotated circuit diagram for the Huruma Estate scenario, labelling all quantities (EMF, internal resistance, terminal voltage, branch currents, power dissipated) and proposing design improvements.	Students synthesised all sub-strand concepts into a unified Final Explanation: the Huruma Estate dim lights and slow kettle at power restoration result from (1) high current demand across many parallel household branches causing significant 'lost volts' (Ir) across the transformer and distribution wiring's internal/line resistance, reducing terminal voltage below 240 V; (2) reduced terminal voltage lowering power delivered to appliances ($P = V^2/R$); (3) undersized or aged distribution cables (high $\rho L/A$) worsening the voltage drop. Design solutions include: using thicker, shorter copper conductors to minimise line resistance; correct fuse/circuit breaker ratings per branch; staged load restoration after blackouts; and each home's circuits being independently parallel-wired so one appliance's failure does not affect others.	Pedagogical note: This capstone lesson is an assessment opportunity. Each group should present their final annotated model — evaluate using a rubric that checks: correct use of $V = IR$, $P = IV$, $Q = I^2Rt$, $R = \rho L/A$, $V_{terminal} = \varepsilon - Ir$, KCL, KVL, and safety device justification. The teacher should resist giving the 'answer' — instead, use Socratic questioning to guide groups to self-correct. End with a whole-class DQB review: physically cross off or annotate each original question with the lesson that answered it, demonstrating cumulative knowledge construction. Final reflection prompt: 'What would you advise a Kenyan electrical engineer tasked with upgrading the Huruma Estate distribution network?' This connects to career awareness (PCIs) and the NGSS practice of communicating findings.